

A Multi-Dimensional, Per-Pass Empirical Study of the LLVM Optimization Pipeline

Ph.D. Student's Activity Report



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22 June 2026

Outline

1 Introduction

2 Math and code

3 Wrap-up

Optimizing compilers

Since the dawn of computing, compiler optimizations have long bridged high-level abstractions and efficient machine code [1–6].

Optimizing compilers [7, 8] use analysis passes to guide transformations [9–11] that enhance performance without altering observable behavior [12–14]—modern architectures (e.g., x86-64, Aarch64) make this an interplay of interdependent passes [9].

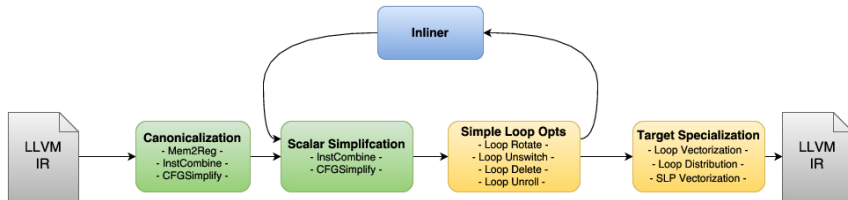
LLVM [15] exemplifies this complexity, serving as the standard backend for C/C++, Rust [16], Julia [17], and via **MLIR** [18]—for *machine learning* (ML) and *deep learning* (DL) stacks [19, 20] that rely on ML/DL optimizing compilers [21–24] (e.g., NVCC¹, OpenXLA [25], TVM [26]) and automotive pipelines.²

¹developer.nvidia.com/cuda-llvm-compiler

²Tesla Full Self-Driving v14.3: stats.tessie.com/versions/2026.2.9.6

LLVM Optimization Pipeline

LLVM is a modular SSA-based compilation framework [27–29]. Its opt tool applies sequences of analysis and transformation passes, forming the core optimization pipeline.



It comprises parameterized nested managers (**module**, **function**, **cgscc**, **loop**, **loop-mssa**), each scoped to a specific IR granularity.

We need better optimization evaluation!

Per-pass impact analysis is critical for compiler engineers and end-users [30–32] to:

- ➡ address *undecidable* [33–35] phase ordering challenges [36–38],
- ➡ improve selection heuristics and ML-based auto-tuning [39–45],
- ➡ guide new optimization design, and
- ➡ understand pass–hardware interactions [34].

However, existing work evaluates LLVM optimizations in isolation or focuses mainly on execution time [46], overlooking multi-dimensional trade-offs—a pass improving runtime may increase binary size or hurt cache locality.

Moreover, the debate over hand-written GPU kernels vs. ML-compiler-optimized kernels remains open [47].³

³octoml-optimizes-apache-tvm-for-apples-m1-beats-core-ml-4-by-29

A Study to Rule Them All (150+ studies at ICSE'26 A*)

We address this gap with a multi-dimensional, systematic empirical study of the LLVM -O3 pipeline, measuring not only **runtime** but also **compile time**, **binary size**, and a broad set of **HW performance counters**⁴ similar to the methodology of Chen *et al.* [51].

RQ₁. *What is the marginal performance impact of individual passes within the LLVM -O3 optimization pipeline?*

RQ₂. *To what extent do optimization passes trade off execution speed, compilation overhead, and binary footprint?*

RQ₃. *How do IR-level transformations affect the evolution of HW performance counters across optimization pipelines?*

RQ₄. *What fraction of achievable speedup is lost to intra-pipeline interference, and which pass drives the loss?*

⁴E.g., those exposed by Model-Specific Registers (MSRs) on x86 CPUs: cache misses, cycles/instructions for ILP, and energy consumption [48–50] via RAPL.

Outline

Paper Structure

Pick and Choose 2mm Kernel from PolyBench

TMP

```
/* D = a*A*B*C + b*D */
for (i = 0; i < NI; i++)
  for (j = 0; j < NJ; j++)
    for (k = 0; k < NK; k++)
      t[i][j] += a*A[i][k]*B[k][j];
for (i = 0; i < NI; i++)
  for (j = 0; j < NL; j++) {
    D[i][j] *= b;
    for (k = 0; k < NJ; k++)
      D[i][j] += t[i][k]*C[k][j];
  }
```

The factorial, set with the Libertine math defaults:

$$fact(n) = \begin{cases} 1 & \text{if } n = 0 \\ n \cdot fact(n - 1) & \text{if } n > 0 \end{cases}$$

Code listings

Monospace is set in Inconsolata:

```
int fact(int n) {  
    if (n == 0) return 1;  
    return n * fact(n - 1);  
}
```

Summary

- ▶ One theme file: `beamerthemescrbe.sty`
- ▶ Same fonts, colors, and ornament as `scribe.cls`
- ▶ Toggle comments with `\scribeshowcommentsfalse`

Thank you!



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